Biochemistry Cumulative Examination

Oct 22, 1999

1. There has been much attention paid in the scientific and popular press to the Human Genome Project and to the sequencing of genomes of other organisms. Simply knowing how to sequence DNA is not sufficient to produce a complete, ordered, and annotated genomic sequence. Suppose you were asked to determine the genome of a new microorganism. Suppose further that you have an adequate supply of genomic DNA, materials, instruments, equipment, and laboratory help. Explain the steps you would go through to get a complete (or very nearly complete) genome sequence. Do *not* explain how DNA sequencing itself works. Concentrate your answer on questions such as how you decide which parts of the genome to sequence and when, how you produce the DNA fragments you sequence, how you assemble smaller bits of sequence into a genome, how you identify interesting features, and other overarching questions that face a genome sequencing project. (30 points)

2. Getting a new genome sequenced is very expensive. Given that we already have a number of genomes, in general what are the expected benefits, if any, of sequencing yet another generic bacterial genome? More specifically, suggest a particular microorganism that you feel should be sequenced (that to the best of your knowledge has not been sequenced) and justify this effort. (30 points)

3. To within one order of magnitude how many proteins are encoded by the human genome? How many in *E. coli*? What percentage of the proteins in *E. coli* have a known function? (10 points)

4. Very little of the DNA in the human genome actually codes for proteins, tRNA, or rRNA. What purpose is served by sequencing this "junk DNA?" (30 points)